

# SYNOPTIC PATTERNS ASSOCIATED WITH HIGH AIR POLLUTION EPISODES IN MEXICO CITY (2012–2024)

*Week 5 – Atmospheric circulation and pollutant anomalies*

*How do large-scale synoptic patterns influence extreme air-pollution days in Mexico City?*

Composite analysis using NARR reanalysis and  
ground-level pollutant data ( $\text{PM}_{2.5}$ ,  $\text{PM}_{10}$ ,  $\text{O}_3$ ,  $\text{NO}_2$ ,  $\text{SO}_2$ )



# WHY STUDY MEXICO CITY'S SYNOPTIC POLLUTION PATTERNS?

- Mexico City (MCMA) = one of the world's most polluted megacities.
- Strong topographic confinement + frequent thermal inversions → pollutant accumulation.
- Previous studies (e.g., Silva-Quiroz et al., 2019; Díaz-Esteban et al., 2022) show the importance of weather-related pollution events.
- Aim: to connect local air-quality extremes with regional atmospheric circulation (500 / 850 hPa).

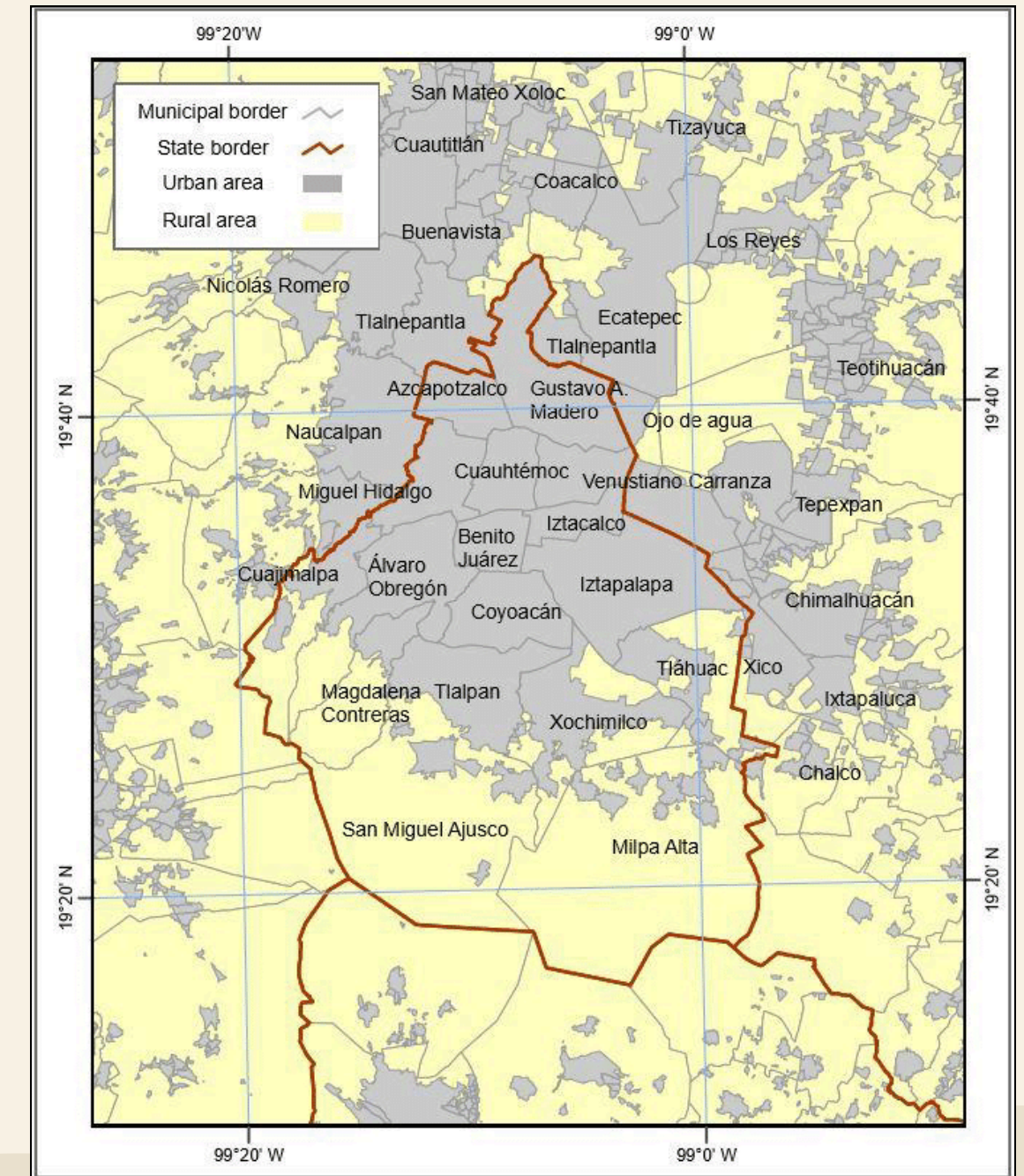
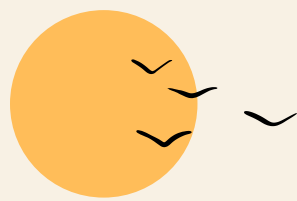


Fig. 1. Mexico City & Metropolitan Area of the Valley of Mexico (based on data from INEGI 2016). (Wieschalla M. N., 2016)



# DATA SOURCES AND COMPOSITE-ANALYSIS WORKFLOW

## Datasets used

- *.nc* NOAA-NARR files (2012–2024): 500/ 850 hPa geopotential heights (HGT) and winds (U,V).
- *.csv* IAYS pollutants file (2012–2024): daily PM<sub>2.5</sub>, PM<sub>10</sub>, O<sub>3</sub>, NO<sub>2</sub>, SO<sub>2</sub> for each station across the city (cleaned, manipulated and transformed for Week 2)

## Workflow

1. Compute daily 90th percentile (p90) for each pollutant.
2. Define “event days” = days > p90.
3. Plot distribution of daily concentrations (p90).
4. Extract NARR fields for the event dates.
5. Calculate anomalies (event – climatology).
6. Test significance ( $p < 0.05$ , t-test).
7. Plot HGT' (shading) + HGT (contours) + winds (vectors).



# DAILY POLLUTANT DISTRIBUTIONS AND P90 THRESHOLDS

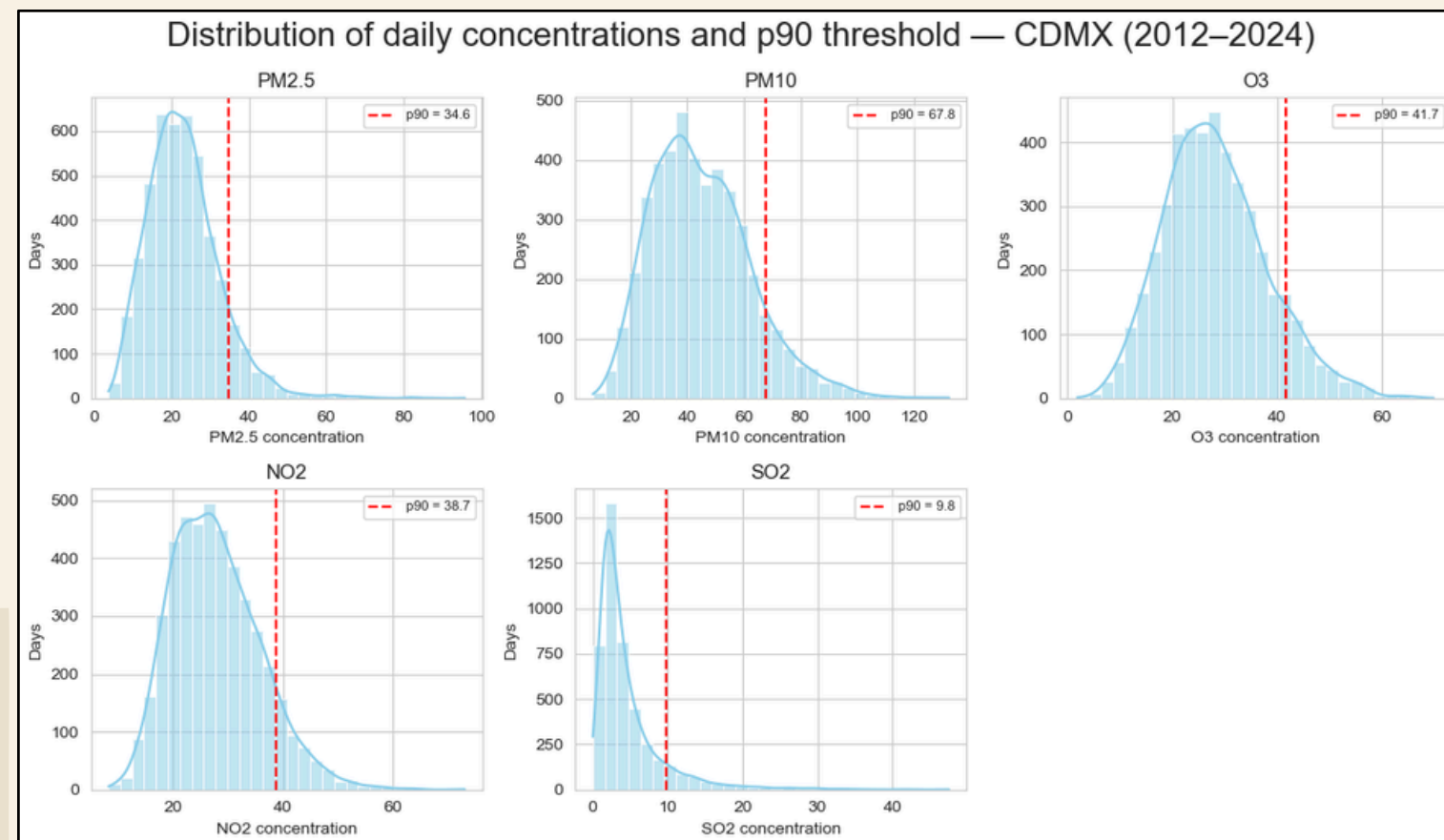


Fig. 3. Daily pollutant distributions (2012–2024) with 90th-percentile thresholds ( $p_{90}$ , red dashed line) defining high-pollution event days used in the composite analysis. (Created by the author).

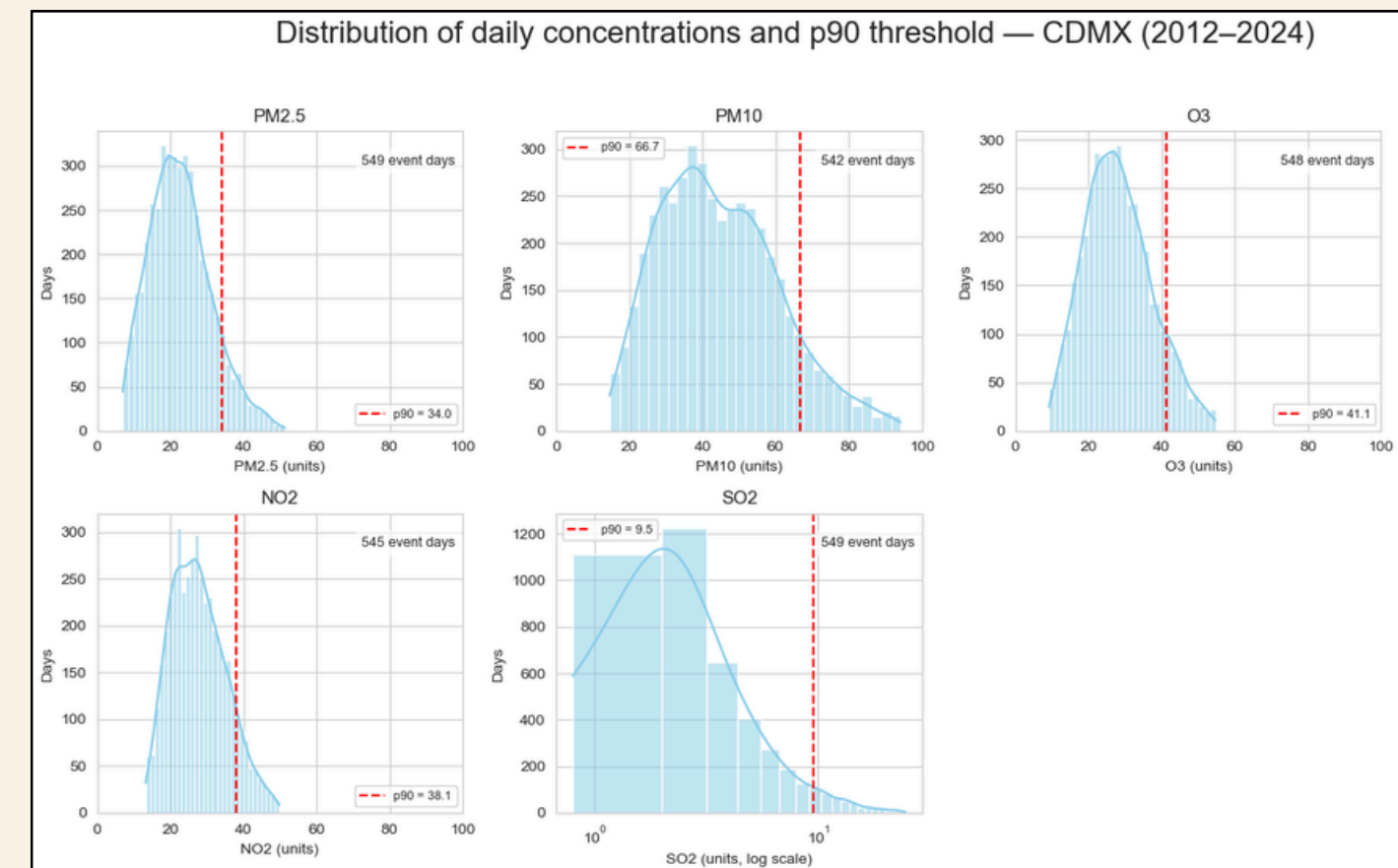
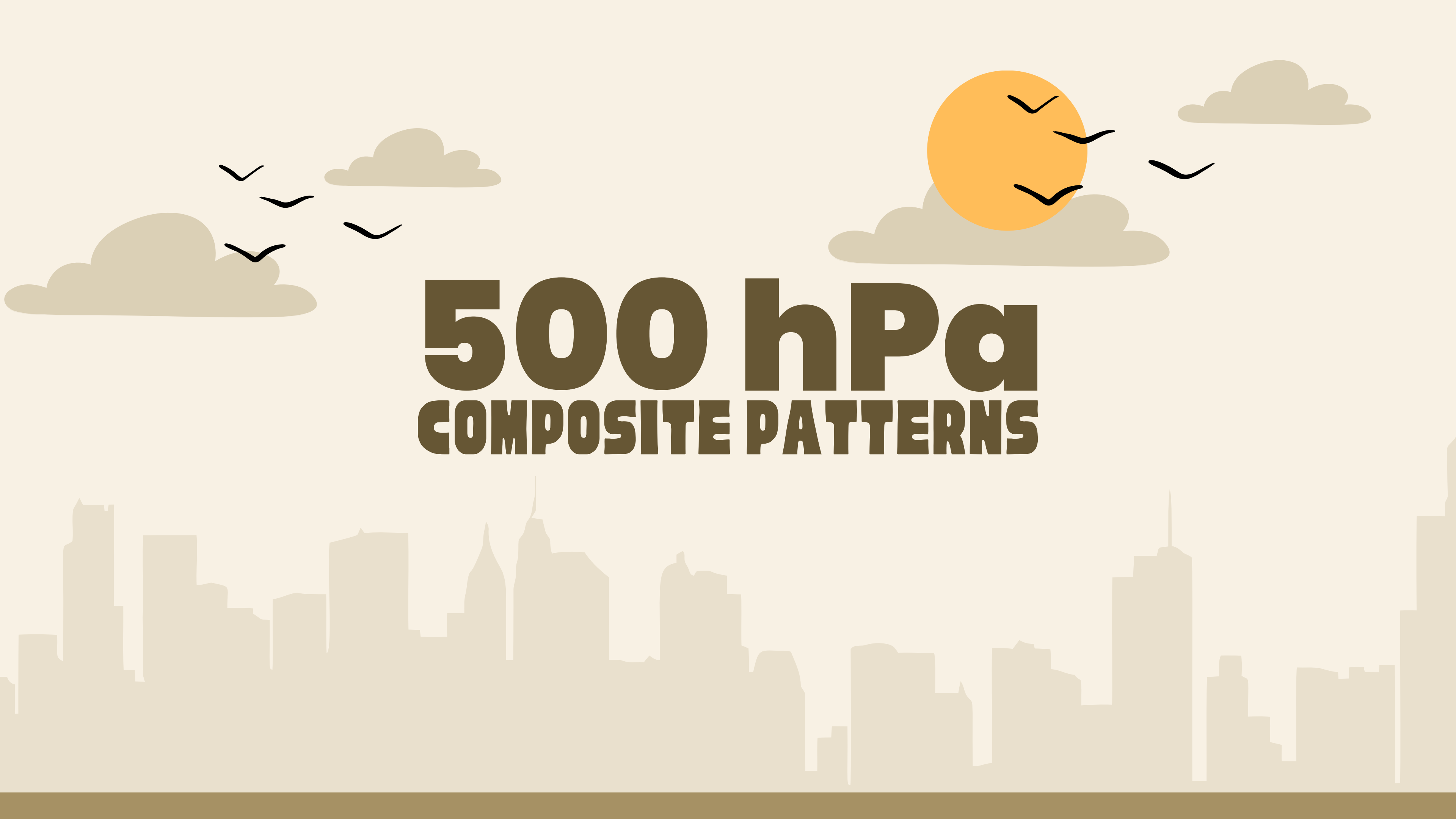


Fig. 4. Daily pollutant distributions (2012–2024) with 90th-percentile thresholds ( $p_{90}$ , red dashed line) defining high-pollution event days used in the composite analysis, with uniform and standardized x-axis scale for PM2.5, PM10, O3 & NO2, and logarithmic scale for SO2. (Created by the author).

- Distributions are right-skewed (typical of air-quality data).
- $p_{90}$  values define ~10 % of days as “high-pollution events.”
- SO<sub>2</sub> requires log scale due to very low baseline levels.

*For each pollutant, the 90th monthly percentile (red line) was identified, used as a threshold for extreme events. These thresholds identify the strongest pollution episodes used in the synoptic composites.*





# **500 hPa**

## **COMPOSITE PATTERNS**



# MID-TROPOSPHERIC (500 HPA) ANOMALIES DURING POLLUTED DAYS (1/3)

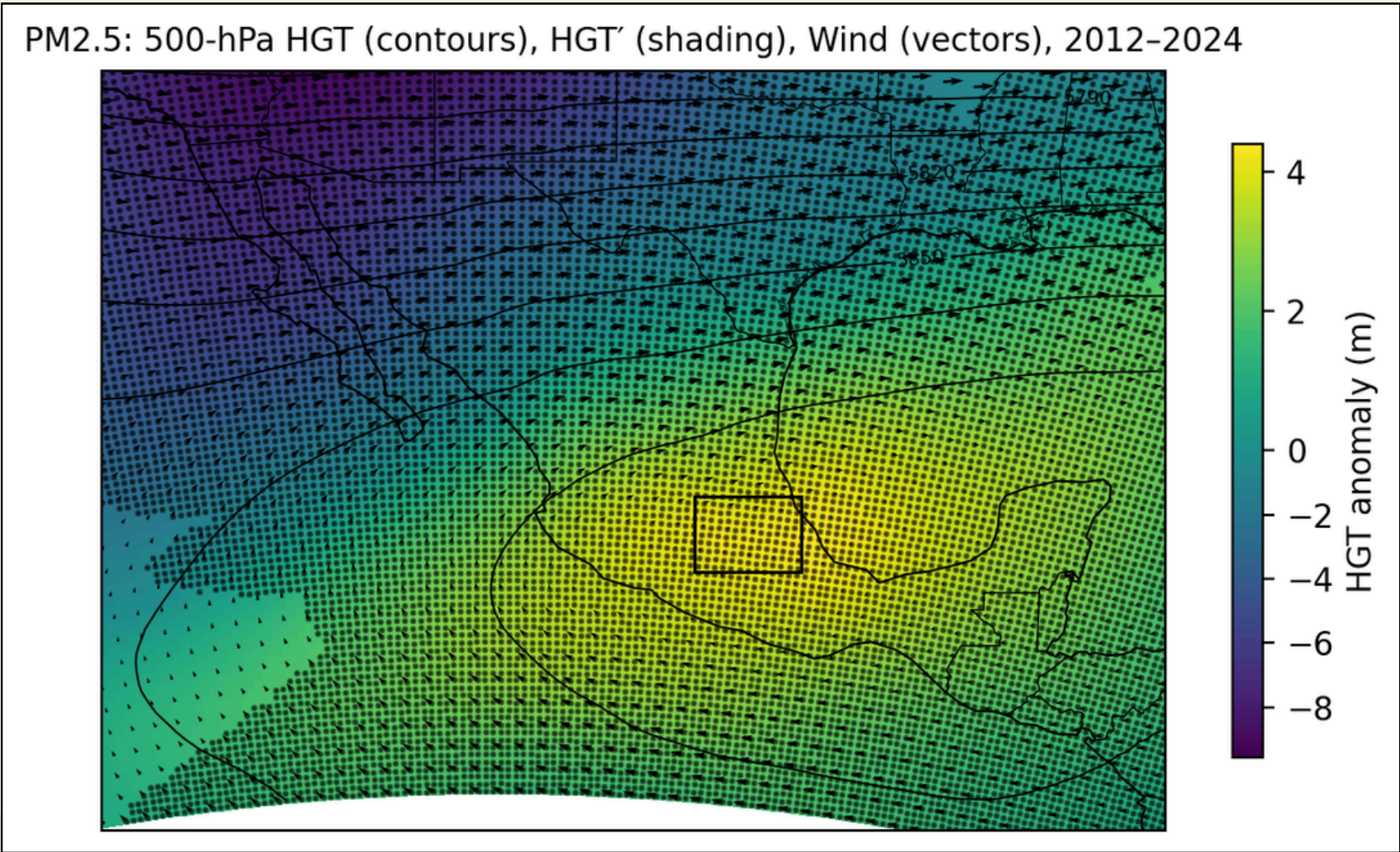


Fig. 5. Composite anomalies of daily mean 500-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 500-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012-2024) for  $\text{PM}_{2.5}$ . The black rectangle marks the MCMA region. Positive HGT' over the high plateau; light average winds  $\rightarrow$  stability and poor ventilation. Compatible with winter events and dry conditions. (Created by the author).

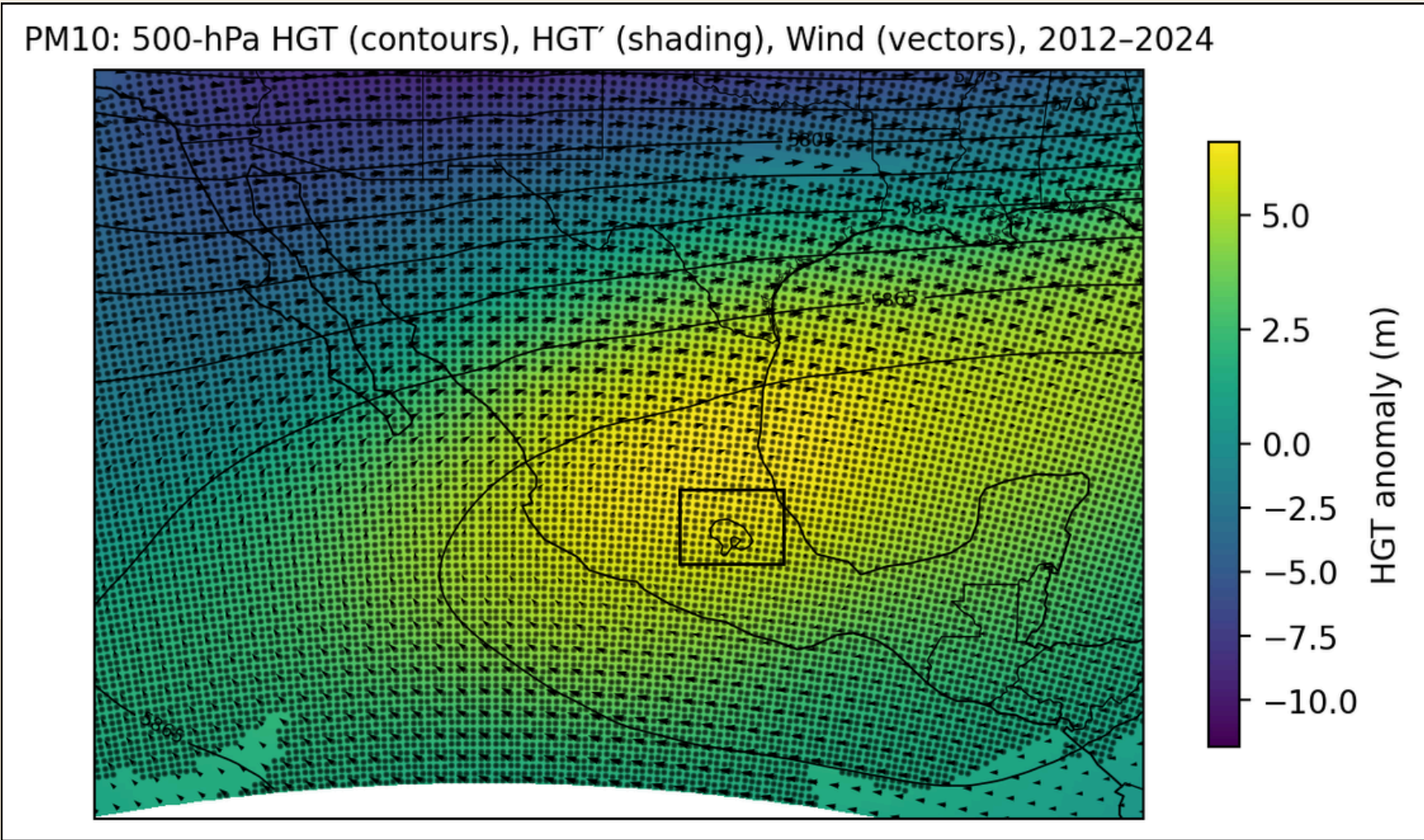


Fig. 6. Composite anomalies of daily mean 500-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 500-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012-2024) for  $\text{PM}_{10}$ . The black rectangle marks the MCMA region. Positive HGT' over the high plateau; light average winds  $\rightarrow$  stability and poor ventilation. Compatible with winter events and dry conditions. In addition, the effect of stippling (significance points  $p < 0.05$ ) can be observed superimposed on a small area; since each grid point is dotted, it looks like a "smear"(Created by the author).



# MID-TROPOSPHERIC (500 HPA) ANOMALIES DURING POLLUTED DAYS (2/3)



O3: 500-hPa HGT (contours), HGT' (shading), Wind (vectors), 2012-2024

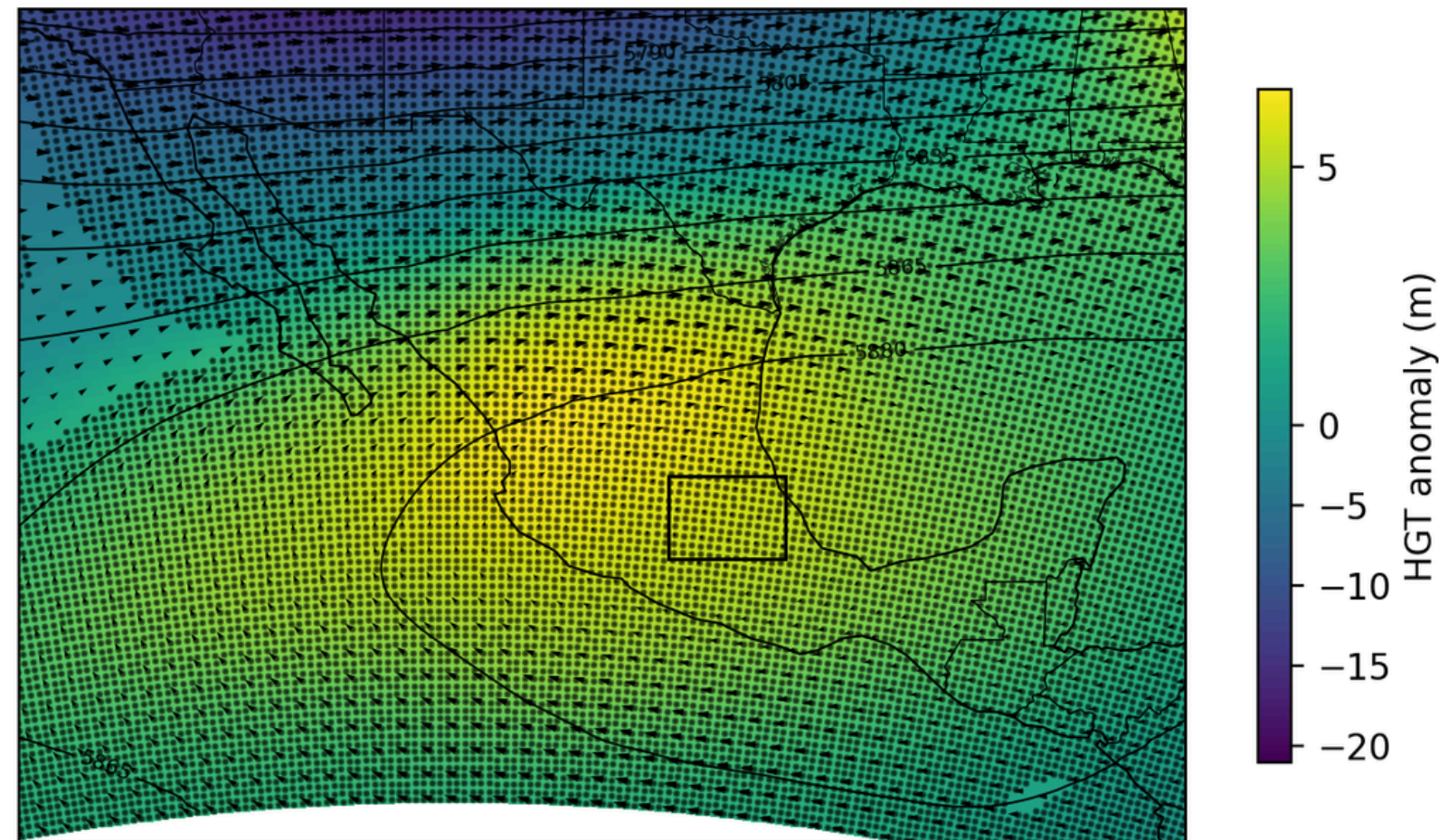


Fig. 7. Composite anomalies of daily mean 500-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 500-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012-2024) for  $\text{O}_3$ . The black rectangle marks the MCMA region. Also HGT' positive; clear sky pattern/subsidence and moderate W-NW / E-SE winds  $\rightarrow$  favored photochemistry and regional transport. Core slightly displaced towards the Pacific. (Created by the author).

NO2: 500-hPa HGT (contours), HGT' (shading), Wind (vectors), 2012-2024

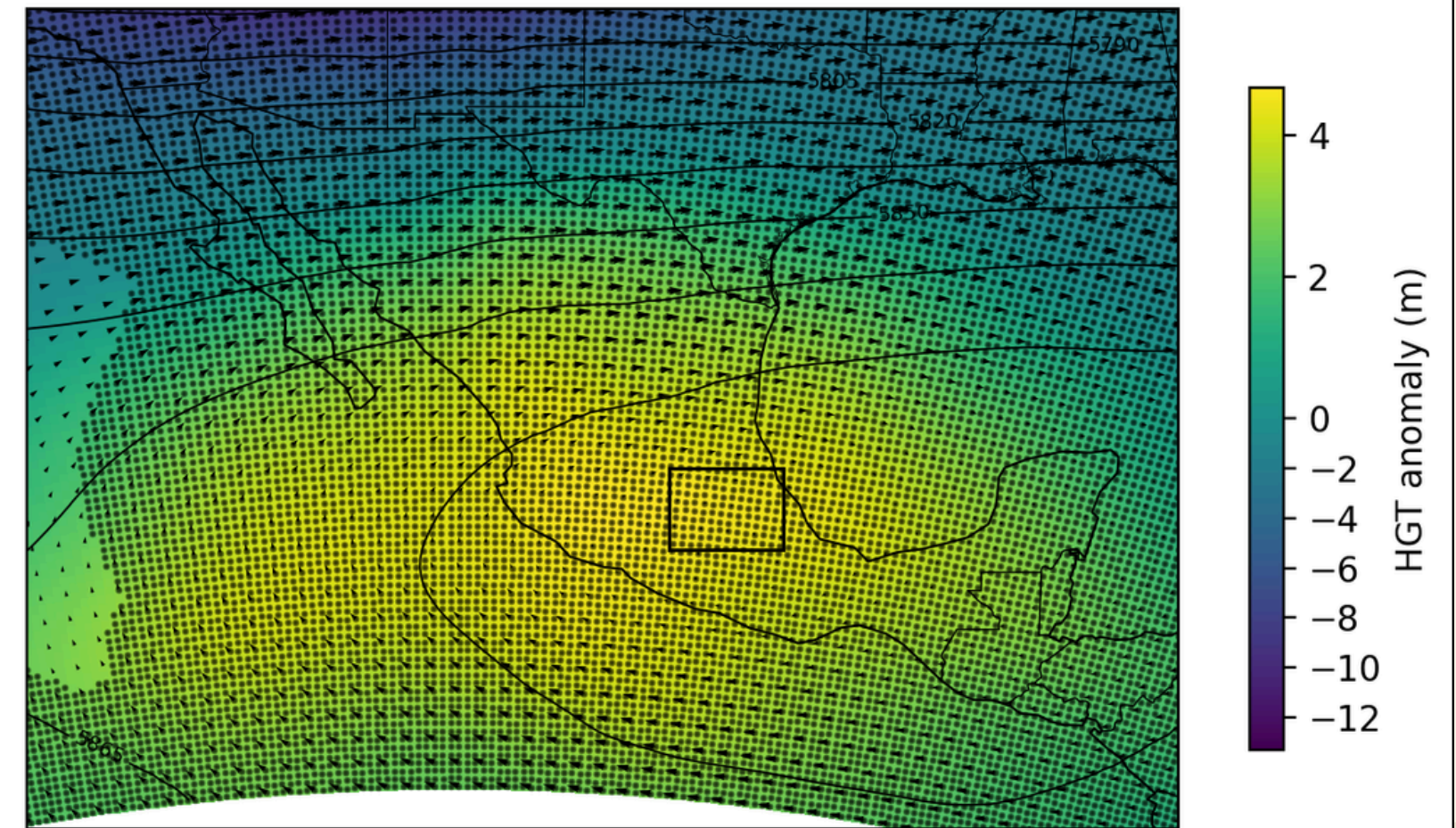


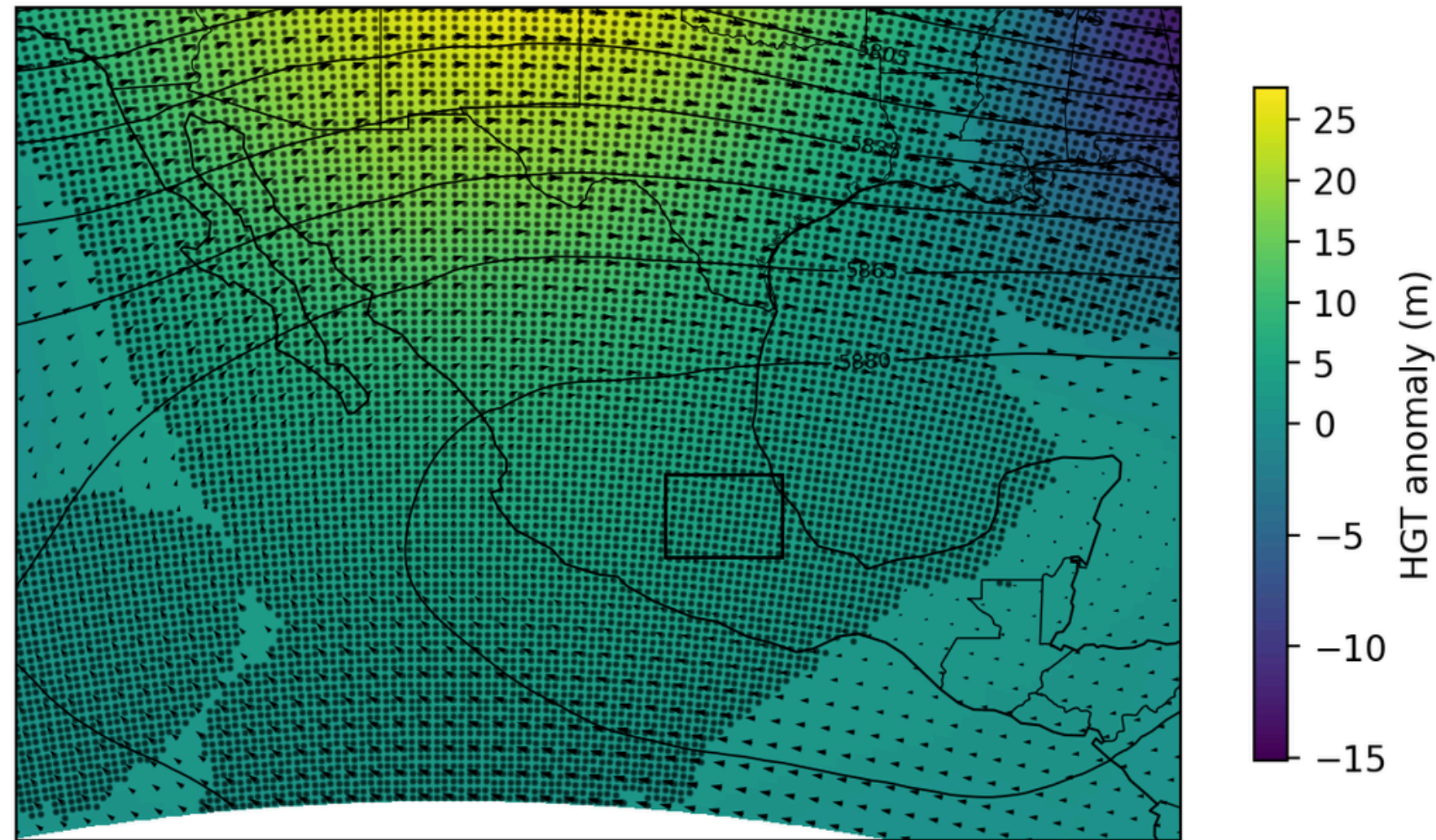
Fig. 8. Composite anomalies of daily mean 500-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 500-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012-2024) for  $\text{NO}_2$ . The black rectangle marks the MCMA region. Pattern very similar to PMs, sign of stagnation and limited vertical mixing. (Created by the author).



# MID-TROPOSPHERIC (500 HPA) ANOMALIES DURING POLLUTED DAYS (3/3)



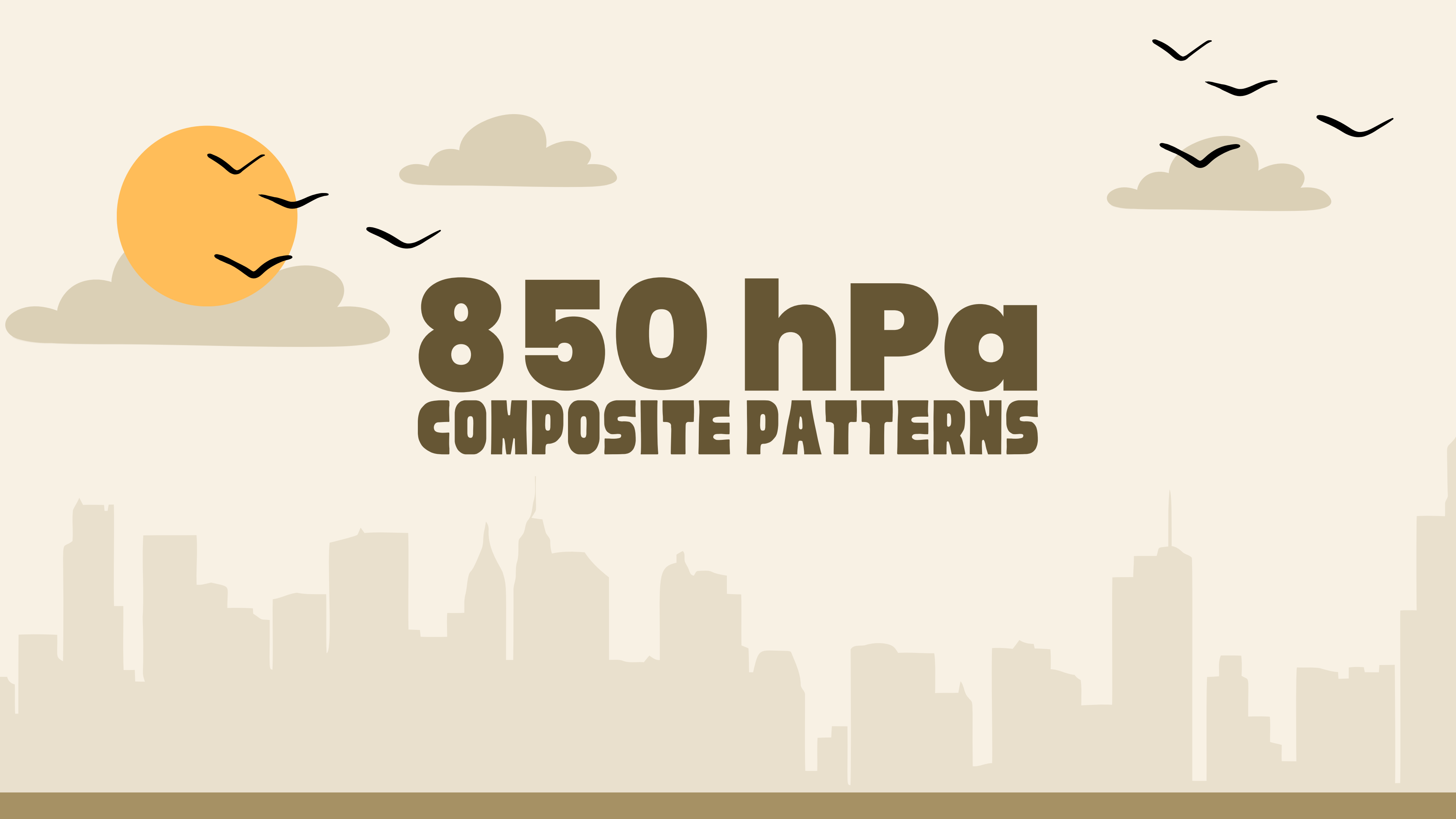
SO<sub>2</sub>: 500-hPa HGT (contours), HGT' (shading), Wind (vectors), 2012-2024



- Widespread positive HGT' (geopotential height anomalies) over central Mexico.
- Indicates subsidence and stagnant conditions, reducing vertical mixing.
- Weak mid-level winds → less ventilation → pollutant build-up.

*High PM and O<sub>3</sub> episodes coincide with regional high-pressure ridges at 500 hPa.*

Fig. 9. Composite anomalies of daily mean 500-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 500-hPa wind vectors (m s<sup>-1</sup>) during high-pollution days in Mexico City (2012-2024) for SO<sub>2</sub>. The black rectangle marks the MCMA region. Slightly positive but still similar signal (Ridge), with a core somewhat displaced towards the Gulf in comparison with O<sub>3</sub>. Additionally, knowing that SO<sub>2</sub> is more source-dependent, synoptic coherence may be slightly lower. (Created by the author).



# 850 hPa

## COMPOSITE PATTERNS



# LOW-LEVEL (850 HPA) CIRCULATION DURING POLLUTED DAYS (1/3)

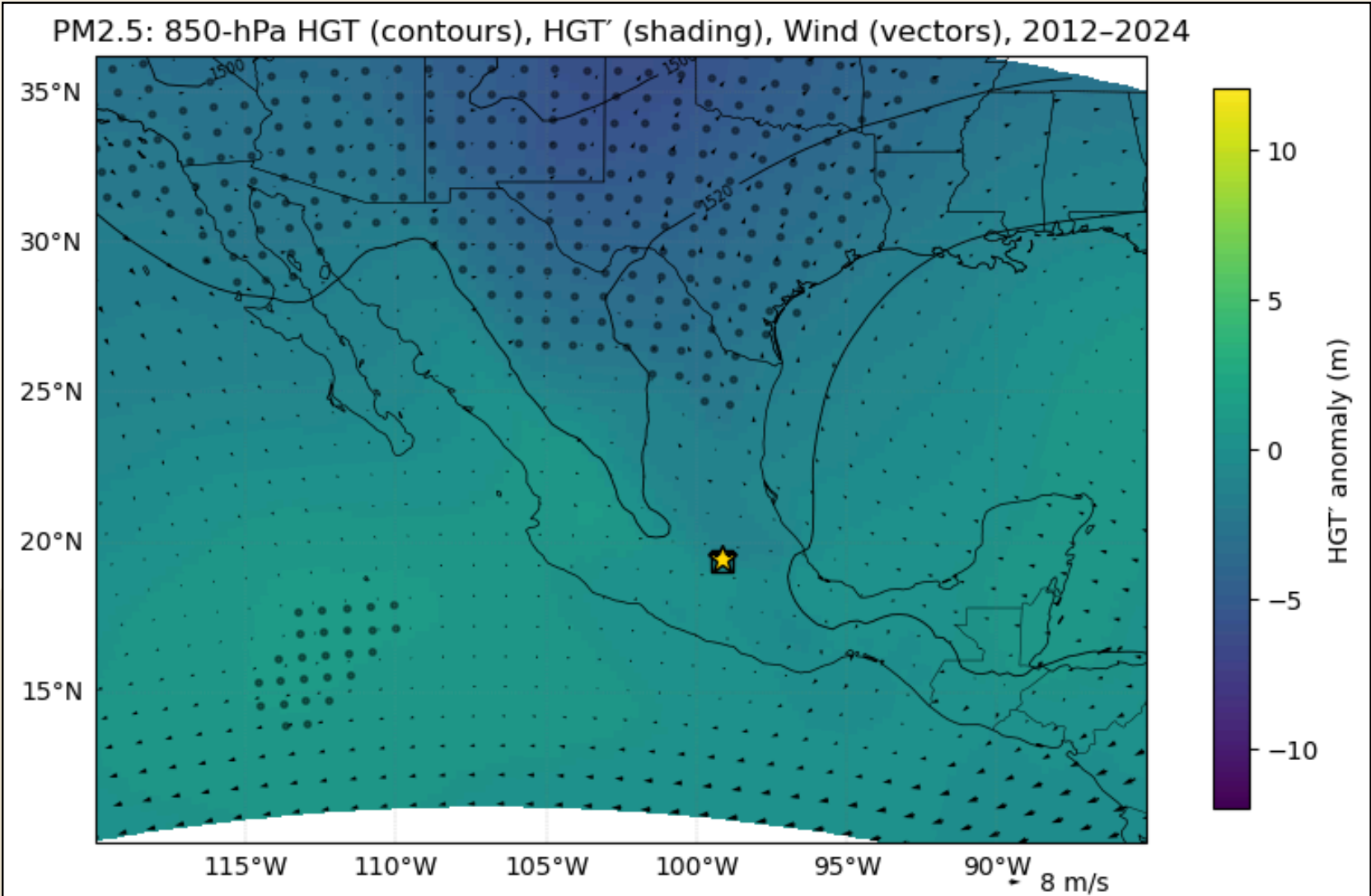


Fig. 10. Composite anomalies of daily mean 850-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 850-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012–2024) for  $\text{PM}_{2.5}$ . The black rectangle marks the MCMA region. The yellow star indicates the location of Mexico City. HGT' (shaded) slightly positive in the highlands/center; weak winds ( $\leq 8 \text{ m/s}$ ) from the E–SE  $\rightarrow$  stagnation and poor ventilation around the MCMA  $\Rightarrow$  favors particle accumulation. (Created by the author).

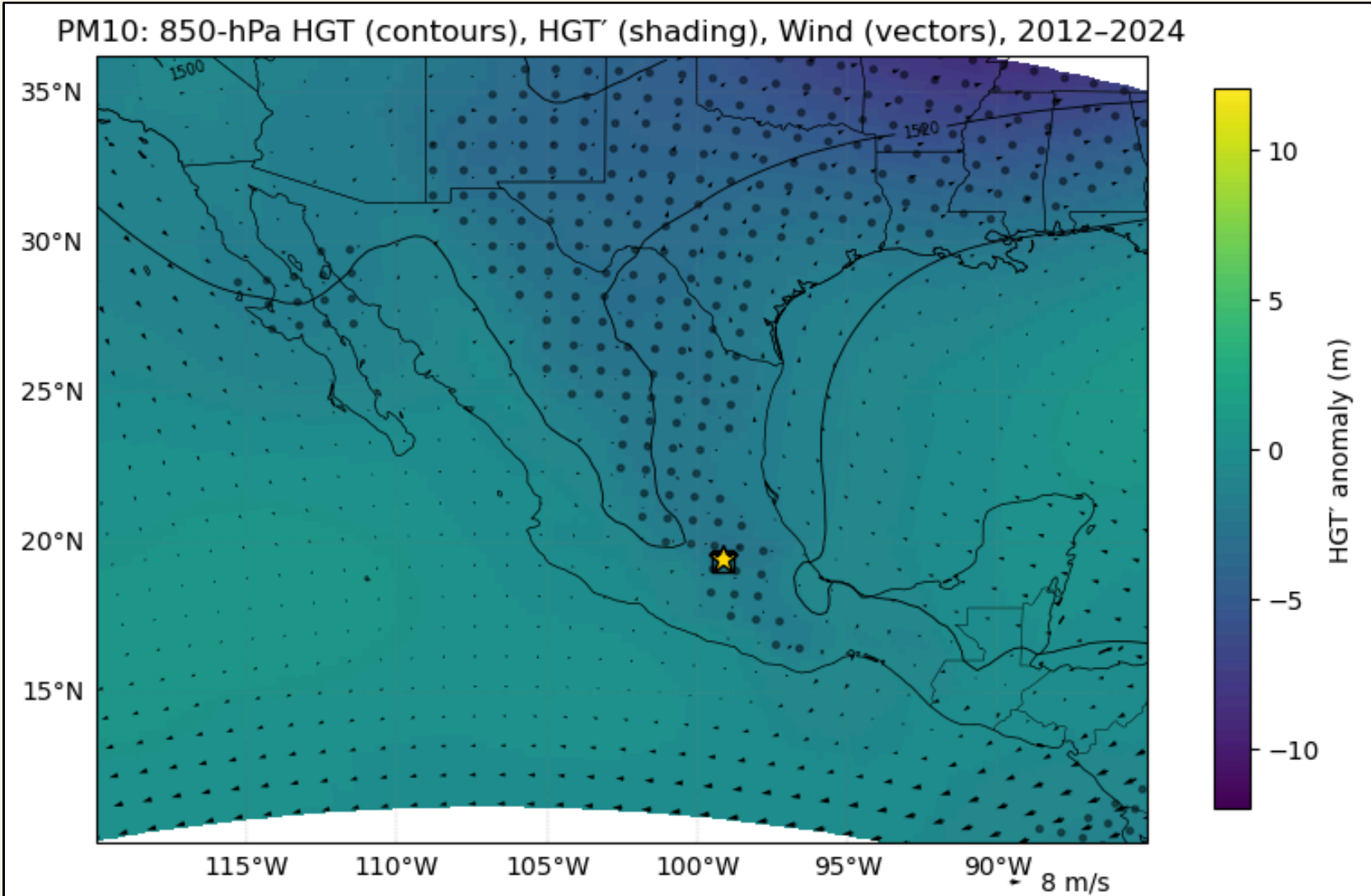


Fig. 11. Composite anomalies of daily mean 850-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 850-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012–2024) for  $\text{PM}_{10}$ . The black rectangle marks the MCMA region. The yellow star indicates the location of Mexico City. HGT' (shaded) slightly positive in the highlands/center; weak winds ( $\leq 8 \text{ m/s}$ ) from the E–SE  $\rightarrow$  stagnation and poor ventilation around the MCMA  $\Rightarrow$  favors particle accumulation. (Created by the author).

# LOW-LEVEL (850 HPA) CIRCULATION DURING POLLUTED DAYS (2/3)

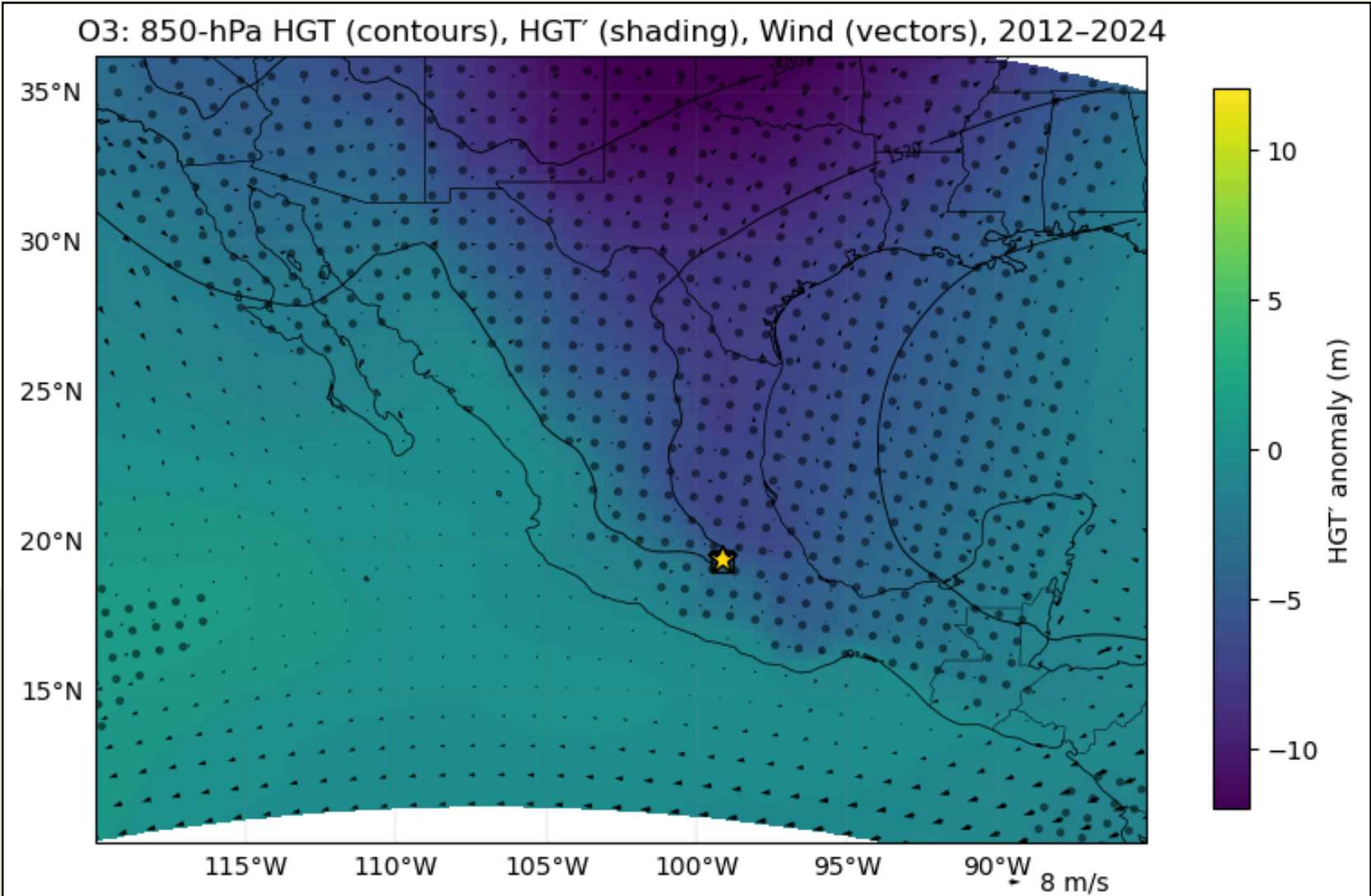


Fig. 12. Composite anomalies of daily mean 850-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 850-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012–2024) for  $\text{O}_3$ . The black rectangle marks the MCMA region. The yellow star indicates the location of Mexico City. The anomaly is somewhat more neutral/mixed in the center, and negative toward the north. However, moderate low flow from the E-SE and regional subsidence: a combination of clear skies and regional transport  $\rightarrow$  favorable conditions for photochemistry. (Created by the author).

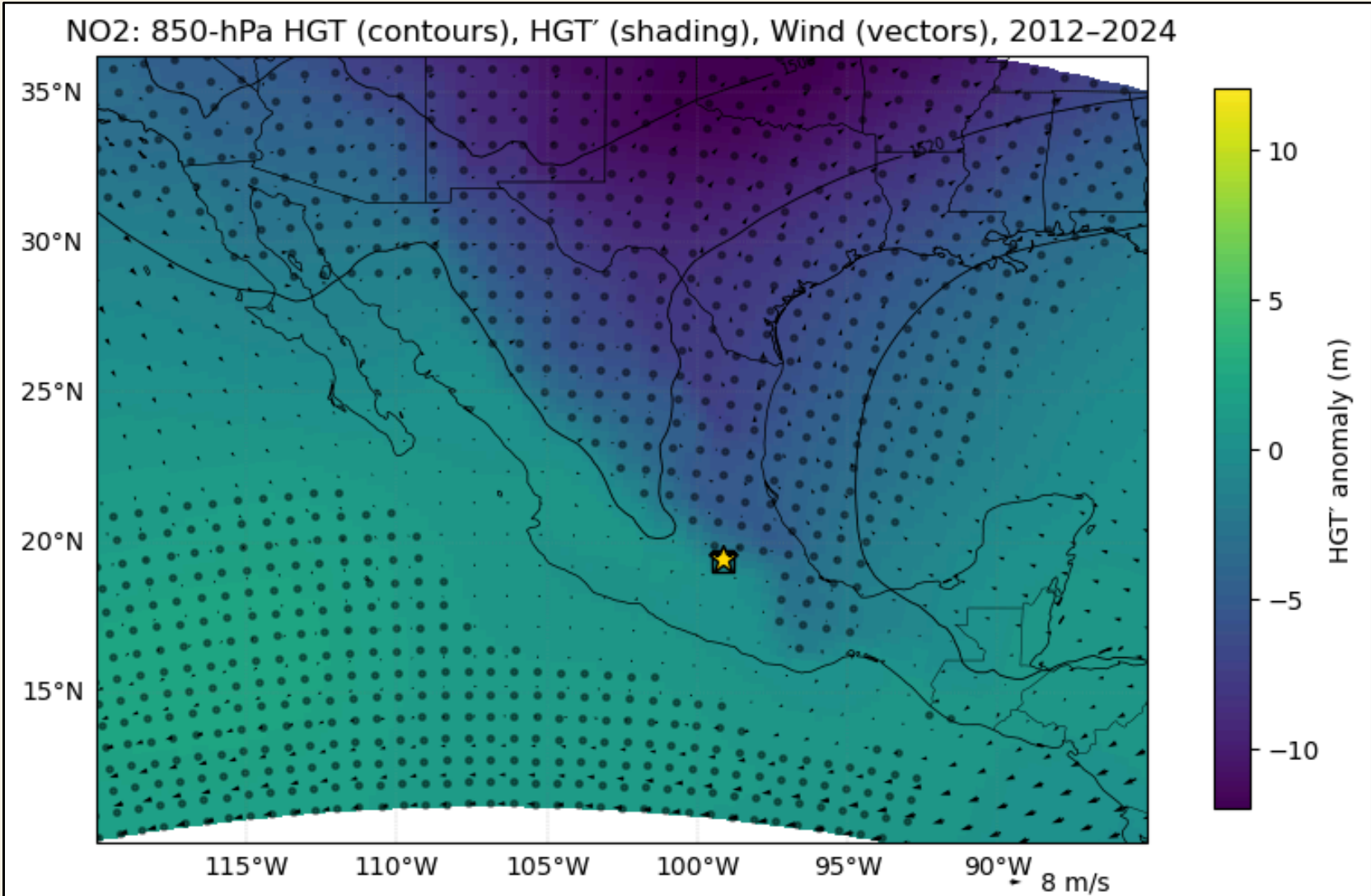


Fig. 13. Composite anomalies of daily mean 850-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 850-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012–2024) for  $\text{NO}_2$ . The black rectangle marks the MCMA region. The yellow star indicates the location of Mexico City. Pattern somewhat similar to PMs and  $\text{O}_3$ : sort of mixed / soft positive HGT' and low wind  $\rightarrow$  stagnation and little vertical mixing (consistent with retained local emissions). (Created by the author).



# LOW-LEVEL (850 HPA) CIRCULATION DURING POLLUTED DAYS (3/3)

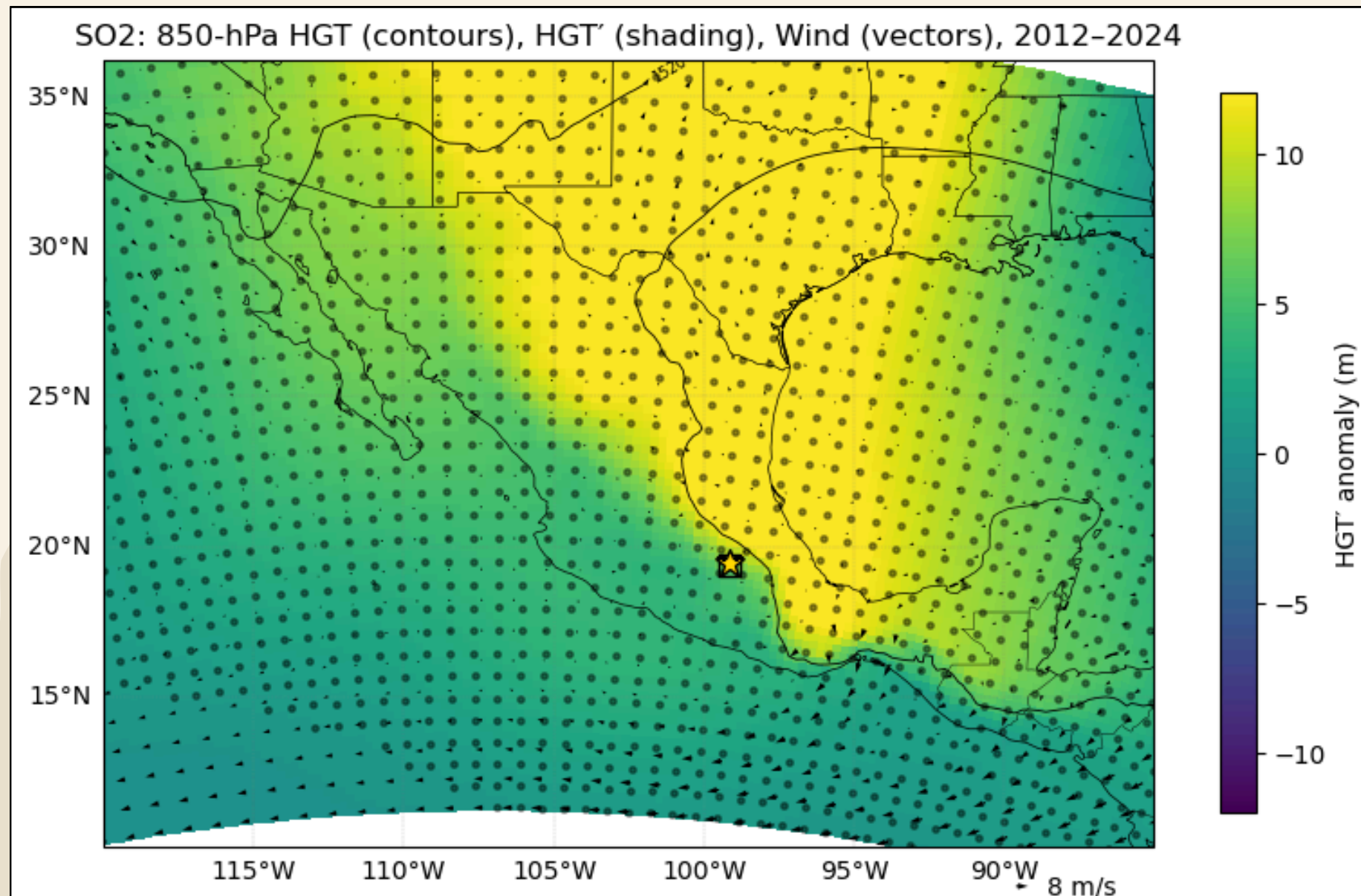
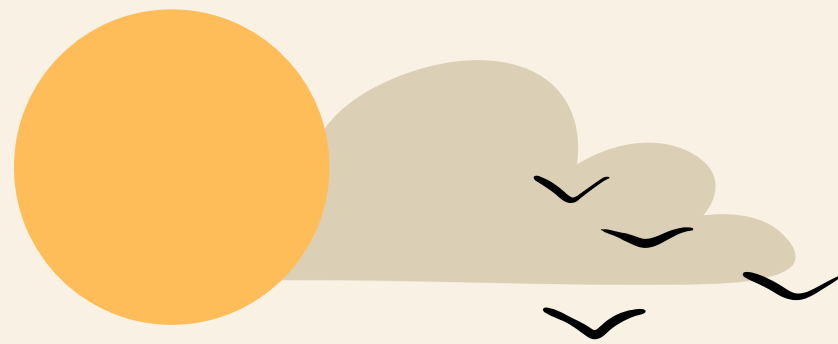


Fig. 14. Composite anomalies of daily mean 850-hPa geopotential height (contours, m), geopotential height anomalies HGT' (shading, m), and 850-hPa wind vectors ( $\text{m s}^{-1}$ ) during high-pollution days in Mexico City (2012-2024) for SO<sub>2</sub>. The black rectangle marks the MCMA region. The yellow star indicates the location of Mexico City. HGT' is more positive and widespread (core toward the Gulf/east of MX) with low-moderate winds. It has a synoptic signal, but same as with 500 hPa, it may differ. (Created by the author).

- Generally negative HGT' over northern Mexico and positive HGT' near the MCMA, implying weak inflow.
- Light winds over central Mexico confirm stagnation near the surface.
- SO<sub>2</sub> shows distinct pattern → local industrial sources influenced by boundary-layer structure.

*850 hPa patterns emphasize surface-level stagnation and local recirculation favorable for pollutant trapping.*



# MULTIPANEL COMPARISON

*Composite anomalies (850 hPa) – five pollutants*

*Despite chemical differences, high-pollution days share a common synoptic signature.*

- Consistent anomalous ridge centered near the MCMA.
- Similar spatial patterns → shared meteorological drivers.
- $\text{SO}_2$  shows localized differences.

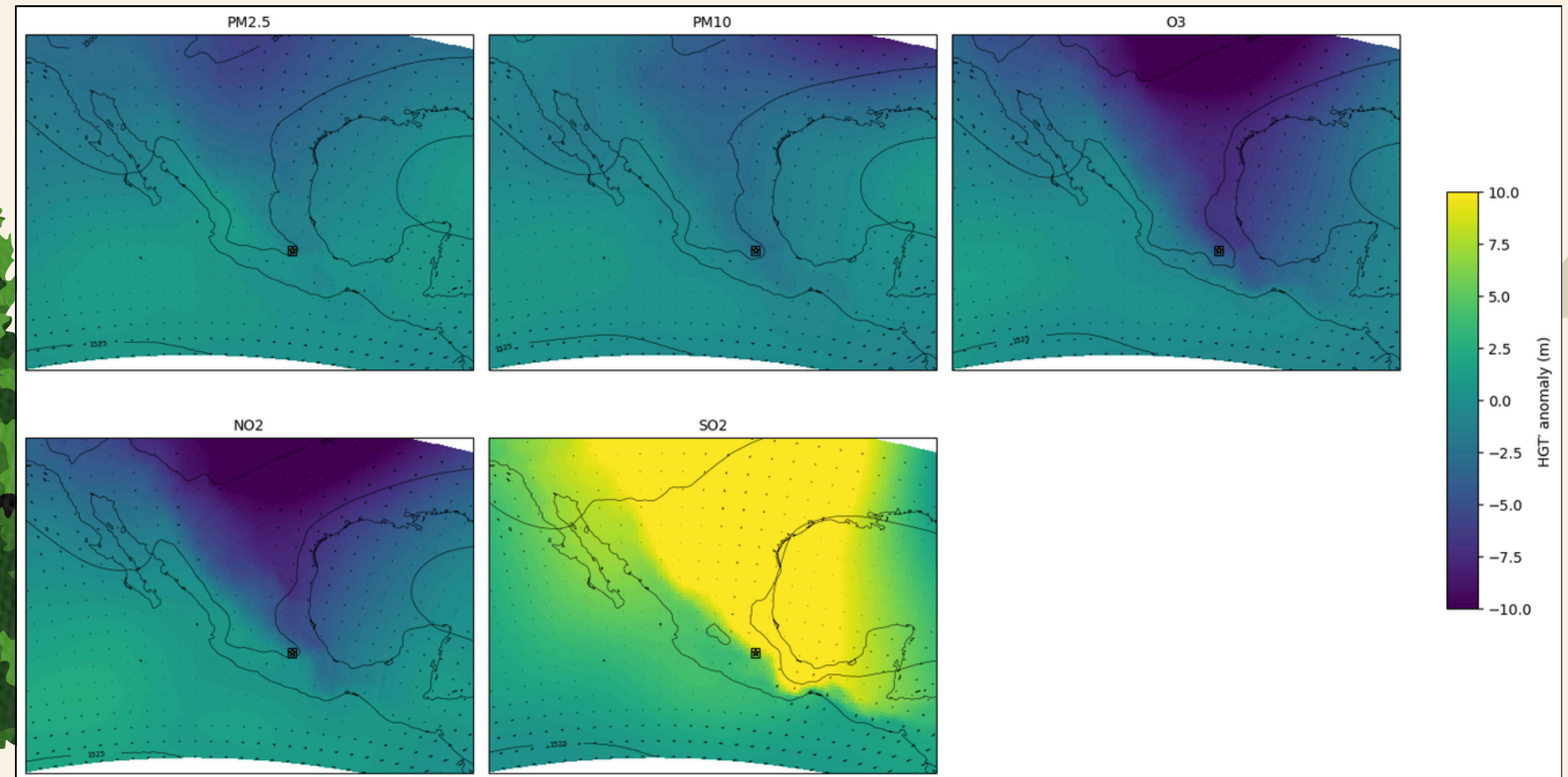


Fig. 15. Composite 850-hPa geopotential height anomalies (HGT', m) and winds ( $\text{m s}^{-1}$ ) for the five major pollutants ( $\text{PM}_{2.5}$ ,  $\text{PM}_{10}$ ,  $\text{O}_3$ ,  $\text{NO}_2$ ,  $\text{SO}_2$ ). Shared color scale shows HGT' in m. Black box denotes MCMA.. Yellow star indicates the location of Mexico City. (Created by the author).





# ASSESSING ROBUSTNESS OF COMPOSITE SIGNALS

*Statistical significance ( $P < 0.05$ )*

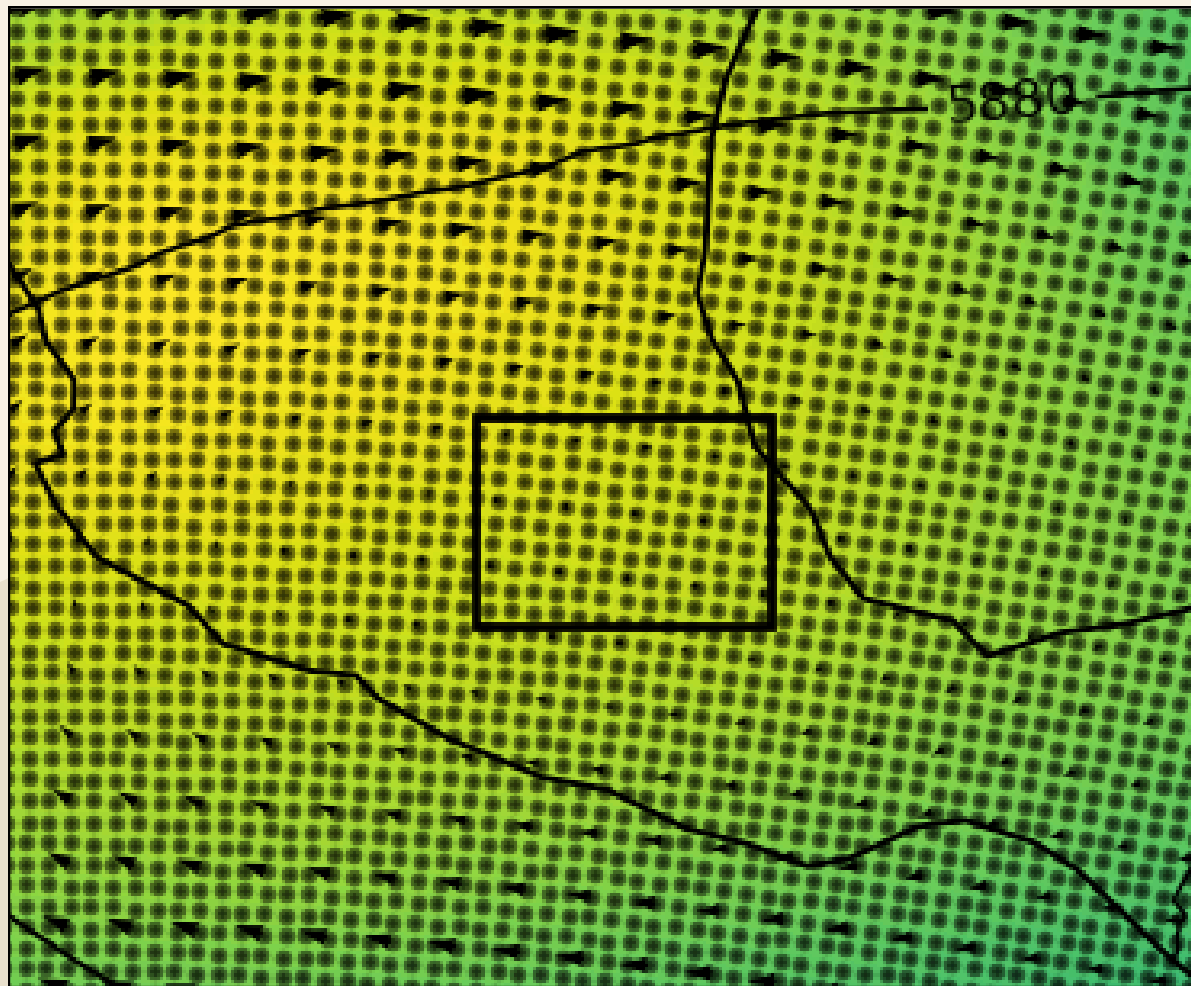


Fig. 16. Zoomed-in 500-hPa composite for  $O_3$ . Stippling/hatched areas denote  $p < 0.05$  (two-sided Student's t-test). (Created by the author).

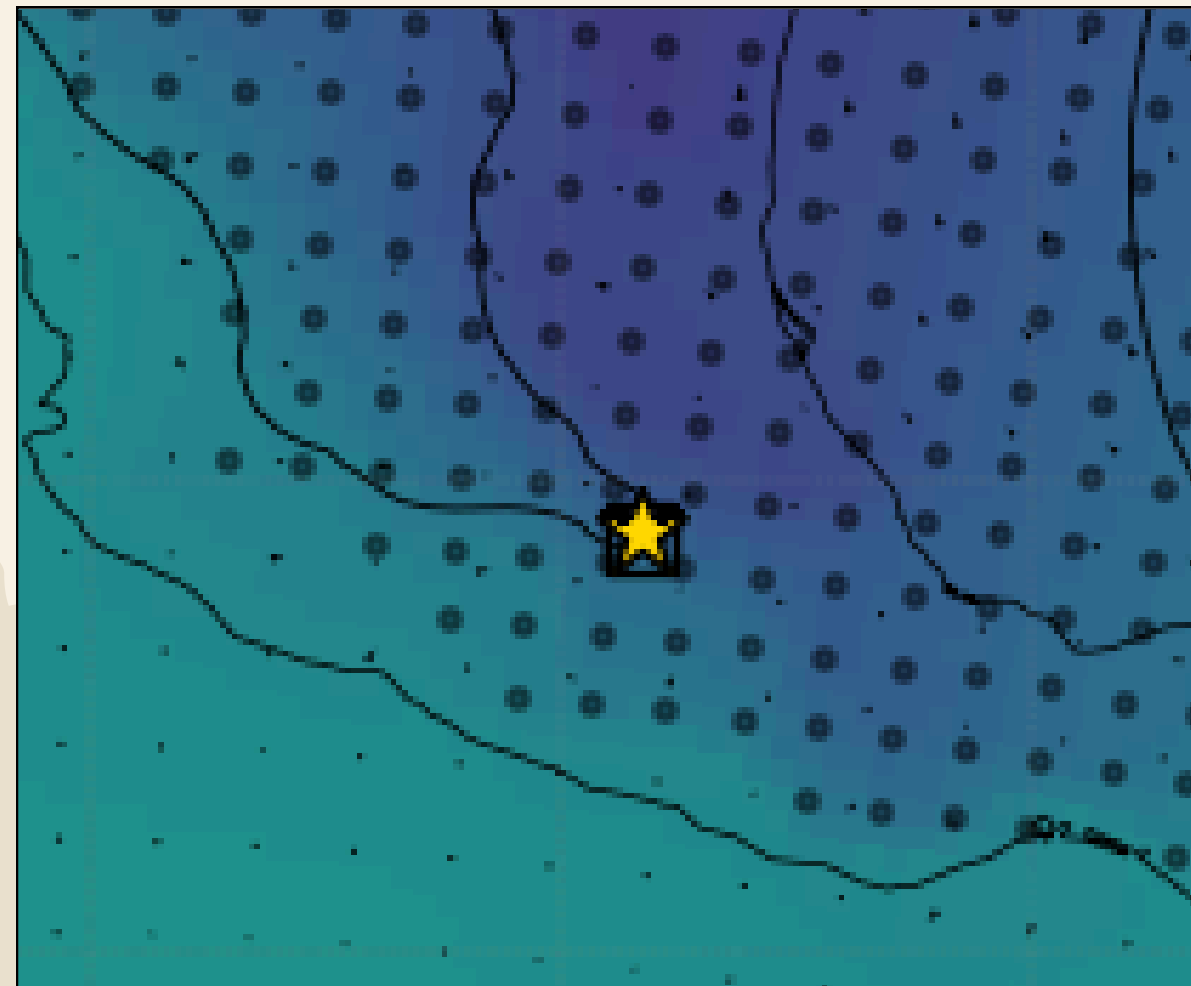
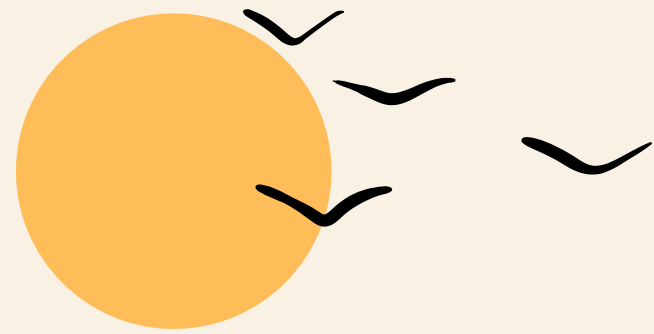


Fig. 17. Zoomed-in 850-hPa composite for  $O_3$ . Stippling/hatched areas denote  $p < 0.05$  (two-sided Student's t-test). (Created by the author).

- Student's t-test used to compare event vs. climatological fields.
- "Stippling" or contour highlight where  $p < 0.05$ .
- Significant regions mainly over central Mexico, confirming signal robustness.





# DISCUSSION: SYNOPTIC COHERENCE & NEXT STEPS

*Learning from the 500 / 850 hPa analysis*

- **Discussion/Conclusion:**

- High-pollution days share a consistent synoptic pattern, with mid-level ridges and weak low-level winds.
- These circulation features imply stagnation and subsidence that favor pollutant accumulation over the MCMA.
- The consistency across all pollutants supports the concept of synoptic coherence in extreme air-quality episodes.

- **Reflections/Improvements:**

- Dig more into xarray and Cartopy libraries. Learn from practice.
- Code optimization is needed: composite calculations and plotting take several hours to run.
- I plan to refine map aesthetics (stippling, color normalization, zoom boxes, annotations) for Week 6.
- Reading more about atmospheric circulation and air-pollution meteorology to strengthen interpretation.

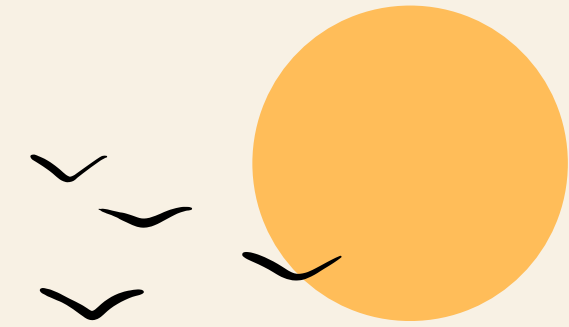
- **Next Steps (week 5 focus + realistic directions):**

- Improve efficiency of NetCDF handling (subsetting and caching).
- Add seasonal separation (dry vs wet) if runtime allows.
- Explore links with near-surface variables (temperature, humidity, PBL height).
- Extend conceptually to Hong Kong using published composites rather than full reanalysis processing.





# REFERENCES



1. Silva-Quiroz R., Rivera A. L., Ordoñez P., Gay-García C., & Frank A. (2019). Atmospheric blockages as trigger of environmental contingencies in Mexico City. *Heliyon*, 5(e02099). <https://doi.org/10.1016/j.scitotenv.2020.143929>
2. Díaz-Esteban Y., Barret B. S., Ordoñez P., & Raga G. B. (2022). Circulation patterns influencing the concentration of pollutants in central Mexico. *Atmospheric Environment*, 274(118976). <https://doi.org/10.1016/j.scitotenv.2020.143929>
3. Wieschalla, M. N. (2016). The concept of the Intercultural City – Opportunities of implementation by key actors in Mexico City. <https://doi.org/10.13140/rg.2.2.36328.47364>
4. NOAA. (n.d.). Physical Sciences Laboratory. <https://downloads.psl.noaa.gov/Datasets/NARR/Dailies/pressure/>
5. Bounding Box Tool: Metadata enrichment for catalogue records by visually selecting geographic coordinates (Latitude / longitude) for maps. (n.d.). <https://boundingbox.klokantech.com/>





# THANK YOU!



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28/10/2025

